GPiBridge: USB-C to IEEE-488 (GPIB)

XC6206: 3.3 V, 200 mA, for the RP2354A only (about 50 mA). Dropout 0.25 V at 100 mA: a USB port sagging to 4.4 V still gives 3.3 V.

Core regulator (1.1 V): Raspberry Pi's parts and layout, copied exactly (gen/rpi_minimal). L1 is polarity-marked: its dot must sit as their footprint shows.

3.3 V and 1.1 V: one 100 nF for every supply pin. On the board each sits at its pin, except where the pin is wailed in by signal lines (pins 6, 11, 20, 23): those join it through the supply rings under the chip.

The transceivers get 5 V only when the firmware sets BUS_PWR_EN, after it has released every line, in reset, crashed or blank, they are off, and TI's parts do not load an unpowered bus. It also keeps 5 V off the RP2354A's pins while it is unpowered: they tolerate 5 V only when powered.

BUS (green): lit whenever the transceivers have power.

CC1, CC2: 5.1 k to ground say 'device': the host then gives 5 V.

USB 2.0 full speed, 12 Mbit/s. 27 R at the chip (RP2350 datasheet).

USB-C. Takes 5 V, up to 270 mA (each transceiver 110 mA max); the transceivers wait for the host's 300 mA.

ABM8-272-T3, 12 MHz, 10 pF load: the crystal Raspberry Pi specifies, with their 15 pF loads and the 1 k that limits its drive.

10 k defaults. With the RP2354A in reset or blank, DC high (device) and TE low (listen): the transceivers cannot drive ATN, REN, IFC or DIO while the 5 V switch opens.

IEEE-488 plug, hand-soldered. Contacts 1–12 run along the wide side of the D. TTL levels, low = true, 0 to 5.5 V; the drivers sink 48 mA and the terminations are inside U2 and U3. Connect only to an IEEE-488 bus.

U2, SN75160B, carries DIO1–8, TE high drives them onto the bus; PE high makes its drivers totem-pole, low open-collector (needed for parallel poll).

U3, SN75161B, carries the control lines. DC low = system controller (drives ATN, REN, IFC; listens to SRQ). TE high = talker (drives DAV, EOI); low = listener (drives NRFD, NDAC).

Raspberry Pi debug connector (JST-SH 3-pin): the Debug Probe's 'D' cable. 100 R on the target side, at the connector (the spec: at the chip). 3.3 V logic.

BOOT: hold it while plugging in to start the RP2354A's USB bootloader. Rarely needed: 'picotool reboot -f -u' gets there from software. RUN stays open: it has its own pull-up inside the RP2354A (datasheet table 1677), and the Debug Probe resets the chip over SWD.

The 2 MB flash is inside the RP2354A: these stay open.

UART0 console, 115200 baud: the Debug Probe's 'U' cable. Works when USB does not.

USB (yellow) and ERR (red) are the firmware's. Both are low-voltage colours, so a 3.3 V pin lights them through 330 R (3 mA).

Direction: the firmware sets TE high before it drives a line, and stops driving before it sets TE low, in one place. The transceivers' terminal outputs are short-circuit limited (15–75 mA).

H1, H2: M2 holes: the two screws that close the housing pass through them.

An RP2354A runs the IEEE-488 handshake and talks to the host over USB as a Prologix-compatible serial controller (and, in a later firmware, a USBTMC/USB488 instrument). Two TI transceivers (SN75160B data, SN75161B control) drive the bus as IEEE 488.1 asks: 48 mA sinks, terminations, hysteresis, and no load on the bus when unpowered. Their 5 V is switched on by the firmware once the host has configured the USB port, so the board is off the bus until the firmware is ready. The plug hangs under the board; everything else is on top.

RP2354A + SN75160B/SN75161B: a Prologix-compatible '++' controller over USB
Hardware CERN-OHL-P-2.0, software MIT
Made by gen/schematic.py from gen/pins.py and gen/parts.py: edit those, not this file
Jason Gallicchio and Claude -- https://github.com/gallicchio/GPiBridge

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